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OPERATIONS RESEARCH

Duality Theory in Linear Programming and the Prices Nobody Sets

Every five minutes a grid operator solves a linear program and the price of electricity falls out of it as a byproduct — a number nobody chose, produced by a theorem from the 1940s about problems that come in mirrored pairs.

2,593 words · 14-minute read

On April 15, 2024, the day-ahead price of electricity in parts of California went negative: generators were paying the grid to accept their output. Three weeks later, on May 8, 2024, prices in Texas ran up against ERCOT's \$5,000/MWh offer cap during a supply crunch [1]. Same market design, same class of software, two numbers four orders of magnitude apart and on opposite sides of zero. Neither was set by a regulator, a trader, or a committee. Both were byproducts — quantities that appear on the side of an optimization problem when a computer finishes solving it, whether or not anyone asks for them.

The grid operator's actual problem is to meet demand at every node of the network at least total cost, subject to generator limits and transmission-line capacities. That is a linear program, re-solved on a five-minute cadence. The prices are not part of its answer; they answer a different question, posed by a second optimization problem that the first generates automatically. The relationship between the two is the duality theory of linear programming, and it is why a number like “the value of a megawatt-hour at this substation right now” can exist without anyone choosing it. The mathematics arrived twice — Leonid Kantorovich's resolving multipliers in Leningrad in 1939 [2], George Dantzig's simplex method at the Pentagon in 1947 [3] — and earned Kantorovich and Tjalling Koopmans a shared Nobel Memorial Prize in 1975 for the economics of allocation that duality makes computable [2], [4].

The Mirror Problem

Write a linear program in its most familiar form. A firm chooses activity levels $x_1, \dots, x_n$ to maximize profit subject to resource limits:

$$\text{maximize } c^T x \quad \text{subject to } Ax \leq b, x \geq 0$$

Here c_j is the profit per unit of activity j , and each row i of $Ax \leq b$ is a resource constraint — machine-hours, crude oil, transmission capacity — with availability b_i . The constraints intersect to form a convex polytope in $\mathbb{R}^n$, and because the objective is linear, an optimum, if one exists, is always attained at a vertex.

Now attach one non-negative variable y_i to each of the m constraints and build a second program from the same data with rows and columns interchanged:

$$\text{minimize } b^T y \quad \text{subject to } A^T y \geq c, y \geq 0$$

This is the dual. Nothing new has been supplied: the same A , b , and c , transposed and read in the other direction. The correspondence is rigid.

Primal (maximize)	Dual (minimize)
n variables $x_j \geq 0$	n constraints $A^T y \geq c$
m constraints $Ax \leq b$	m variables $y_i \geq 0$
objective coefficients c	right-hand sides c
right-hand sides b	objective coefficients b

The claim that makes this more than bookkeeping is that the two problems have the same optimal value. Solve either one and you have solved both.

Farkas' Lemma and the Two Halves of the Theorem

One half is nearly free. For any primal-feasible x and any dual-feasible y ,

$$c^T x \leq (A^T y)^T x = y^T (Ax) \leq y^T b = b^T y,$$

where the first inequality uses $A^T y \geq c$ together with $x \geq 0$, and the second uses $Ax \leq b$ together with $y \geq 0$ [5]. That is the entire proof of weak duality: every dual-feasible solution is an upper bound on every primal-feasible one. It costs two sign arguments, and it already does real work: any feasible y , however obtained, certifies that no primal plan can beat $b^T y$.

Rearranging the same chain gives an identity worth keeping. For feasible x and y , the gap between the two objectives splits into two non-negative pieces:

$$b^T y - c^T x = y^T (b - Ax) + x^T (A^T y - c).$$

Both terms are inner products of non-negative vectors, so each is at least zero, and the gap vanishes only if both vanish term by term. That observation is the whole content of complementary slackness, stated as a theorem: feasible x and y are simultaneously optimal if and only if $y_i(b_i - A_i x) = 0$ for every constraint i and $x_j(A_j^T y - c_j) = 0$ for every variable j [6]. For each matched pair, either the primal constraint is tight or its dual variable is zero — never both slack at once. No appeal to the simplex method is required.

The hard half is strong duality: if the primal has a finite optimum, so does the dual, and the sandwich closes exactly, $\mathbf{c}^T \mathbf{x}^* = \mathbf{b}^T \mathbf{y}^*$ [5]. Weak duality alone cannot deliver this — it forbids the dual from dipping below the primal but says nothing about the two meeting. What closes the gap is a separation theorem about linear inequalities, usually stated as Farkas' lemma: for a matrix $\mathbf{A}$ and vector $\mathbf{b}$, exactly one of the following holds. Either there exists $\mathbf{x} \geq \mathbf{0}$ with $\mathbf{Ax} = \mathbf{b}$, or there exists $\mathbf{y}$ with $\mathbf{A}^T \mathbf{y} \geq \mathbf{0}$ and $\mathbf{b}^T \mathbf{y} < 0$. Geometrically, either $\mathbf{b}$ lies inside the cone spanned by the columns of $\mathbf{A}$, or a hyperplane separates it from that cone — and the separating hyperplane's normal vector is a finite, checkable certificate that no solution exists [5].

That dichotomy converts an absence into an object. Let $\mathbf{v}$ be the primal optimum. Then the system asking for a feasible $\mathbf{x}$ with $\mathbf{c}^T \mathbf{x} > \mathbf{v}$ is infeasible, by definition of $\mathbf{v}$. Farkas' lemma says an infeasible linear system always hands back a certificate, and when you write out the certificate for this particular system, its components are exactly a dual-feasible $\mathbf{y}$ with $\mathbf{b}^T \mathbf{y} \leq \mathbf{v}$. Weak duality gives $\mathbf{b}^T \mathbf{y} \geq \mathbf{v}$. The two bounds pin the value, and strong duality follows. A second, more constructive route runs simplex to termination, where the reduced costs it has tracked all along are already a dual-feasible solution with matching objective, by how the algorithm certifies that no adjacent vertex improves [5], [6].

What a Dual Variable Measures

Units settle the interpretation. If constraint i is denominated in megawatt-hours and the objective in dollars, $\mathbf{y}_i$ must carry dollars per megawatt-hour. It is a price — specifically the shadow price of resource i , the derivative of the optimal value with respect to that resource's availability, $\mathbf{y}_i^* = \partial z^* / \partial b_i$ [6].

This is where duality stops being formal symmetry and becomes an accounting system. A binding constraint's shadow price is the most a rational planner should pay for one more unit of the thing it limits — the break-even number for a debottlenecking project, an incremental purchase, an overtime shift. A slack constraint's shadow price is zero, and complementary slackness guarantees it: an unused resource cannot be limiting profit. Taken together, the dual solution lets a firm price its own internal scarcities as though they were traded, and evaluate any proposed new activity against the imputed cost of the resources it would consume [6], [7]. Nobody sets these prices. They are implied by the constraint set.

A Plant You Can Solve by Hand

A stylized planning problem: two product lines, two internal capacities, one hard cap. This is a standard teaching instance, small enough to solve by inspection and rich enough to show every moving part [8].

$$\begin{aligned} &\textbf{maximize} && z = 3x_1 + 5x_2 \\ &\textbf{subject to} && x_1 \leq 4 \\ &&& 2x_2 \leq 12 \\ &&& 3x_1 + 2x_2 \leq 18 \\ &&& x_1, x_2 \geq 0 \end{aligned}$$

The feasible region is a pentagon with vertices $(0, 0)$, $(4, 0)$, $(4, 3)$, $(2, 6)$, and $(0, 6)$, where the objective takes the values **0, 12, 27, 36**, and **30**. So $\mathbf{x}^* = (2, 6)$ and $\mathbf{z}^* = 36$. Note which constraints bind there: $\mathbf{x}_1 = 2 < 4$ leaves the first with slack of **2**, while $2(6) = 12$ and $3(2) + 2(6) = 18$ are both exactly tight.

Transposing gives the dual — three variables, two constraints, one per primal activity:

$$\begin{aligned} \text{minimize } w &= 4y_1 + 12y_2 + 18y_3 \\ \text{subject to } y_1 + 3y_3 &\geq 3 \\ 2y_2 + 2y_3 &\geq 5 \\ y_1, y_2, y_3 &\geq 0 \end{aligned}$$

There is no need to solve this from scratch. Complementary slackness determines it. The first primal constraint is slack, so $y_1 = 0$. Both primal variables are strictly positive, so both dual constraints hold with equality: $y_1 + 3y_3 = 3$ gives $y_3 = 1$, and $2y_2 + 2y_3 = 5$ gives $y_2 = 1.5$. The dual objective is then $4(0) + 12(1.5) + 18(1) = 18 + 18 = 36$, matching the primal optimum exactly, as strong duality promises.

TRY THIS BEFORE READING ON

Raise the third constraint's right-hand side from **18** to **19**, leaving everything else alone, and re-solve in your head. Then do the same for the second constraint, **12** $\rightarrow$ **13**. How far does the optimum move in each case?

Loosening the third constraint to **19** leaves $\mathbf{x}_2 = 6$ fixed by the second constraint and lets $\mathbf{x}_1$ rise to **7/3**, giving $\mathbf{z} = 3(7/3) + 30 = 37$: a gain of exactly **1**, which is y_3 . Loosening the second constraint to **13** pushes $\mathbf{x}_2$ to **6.5** and squeezes $\mathbf{x}_1$ down to **5/3**, giving $\mathbf{z} = 5 + 32.5 = 37.5$: a gain of exactly **1.5**, which is y_2 . Loosening the first constraint to **5** changes nothing at all, since $\mathbf{x}_1$ was not against its cap; the gain is **0**, which is y_1 . The dual variables were the marginal values all along.

Primal constraint	Binding?	y_i^*	Meaning
$\mathbf{x}_1 \leq 4$	No, slack 2	0	not the bottleneck; another unit is worthless
$2\mathbf{x}_2 \leq 12$	Yes	1.5	another unit adds \$1.50 of optimal profit
$3\mathbf{x}_1 + 2\mathbf{x}_2 \leq 18$	Yes	1	another unit adds \$1.00 of optimal profit

That is the theory in miniature. Constraints that bind get positive prices, the slack one gets zero, and those prices multiplied against the resource endowments reconstruct the same optimal value the primal reached by a completely different route.

The Price at the Node

Now scale the picture up to a continent. The locational marginal price at a grid node — the number that went negative in California and hit the cap in Texas — decomposes into exactly three pieces: an energy component reflecting the system-wide marginal generator's bid, a loss component reflecting resistive losses, and a congestion component computed as a shift factor times the shadow price on a transmission-line capacity constraint [1].

That congestion term is complementary slackness rendered in dollars. While a line has spare capacity, its constraint is slack, its shadow price is zero, and it contributes nothing to any node's price; the instant it binds, the shadow price jumps positive and prices on either side of the constraint separate. This is why one interconnected system can post negative prices in one county and cap-level prices in another at the same moment: surplus generation trapped behind a binding line cannot reach the load that would pay for it, and the LP prices the two locations as if they were nearly separate markets, because for the next marginal megawatt-hour they are. Negative prices need no appeal to irrationality either — when must-run or subsidised output exceeds what the network can carry away, the cheapest way to satisfy a binding constraint is to pay someone to absorb energy.

The standard qualification is that real market clearing is not a pure linear program. Unit commitment — the on/off decisions for thermal generators, with start-up costs and minimum run times — is mixed-integer, and integer programs have no exact duality theory of this kind. Operators fix the commitment decisions and take shadow prices from the remaining continuous dispatch problem, so published prices are approximately, not exactly, the marginal values duality defines; the residue is settled through side payments rather than prices.

Speed matters here, because the clearing runs every five minutes and simplex has no polynomial worst-case guarantee: in 1972 Victor Klee and George Minty built a perturbed hypercube in n dimensions whose 2^n vertices Dantzig's original pivot rule is forced to visit one by one [9], [10]. Yet simplex remains the industrial default, solving problems with millions of variables in seconds [11]. Two results reconciled theory with practice — Narendra Karmarkar's 1984 interior-point method, with a genuine polynomial bound of about $O(n^{3.5} L^2 \log L \log \log L)$ operations [12], and Spielman and Teng's smoothed analysis in 2001, showing that perturbing any input, even an adversarial one, by a little random noise collapses the expected running time to polynomial [13]. Spielman's description of the worst cases is that they are so fragile that "if you are wearing boxing gloves, you couldn't build one of them" [13]. Measured data always wears boxing gloves.

Minimax, Rounding, and the Kernel Trick

Once you can recognise a primal-dual pair, it turns up in places that were never described as resource allocation.

John von Neumann's 1928 minimax theorem says that every two-player zero-sum game has a value, with optimal mixed strategies for both sides. It is LP strong duality in different clothing: the row player's optimization and the column player's are literally a dual pair, the game's value is the common optimal objective, and solving a zero-sum game numerically today means handing it to an LP solver [15], [16]. One player's equilibrium mixed strategy is the shadow-price vector of the other's problem — the same object that prices a transmission line, now describing how often to play rock.

Approximation algorithms lean on the same bound in the other direction. Vertex cover, set cover, and many scheduling problems are NP-hard as integer programs, but their LP relaxations — letting the variables range over $[0, 1]$ instead of $\{0, 1\}$ — solve in polynomial time and bound the integer optimum, since every integer solution is feasible for the relaxation. For vertex cover, solving the relaxation and rounding every variable at least $1/2$ up to 1 yields a cover at most twice the true optimum: a factor-2 guarantee falling out of the LP bound with no cleverness beyond the rounding rule [17].

And the support-vector machines that carried machine learning for a decade before deep networks are convex quadratic programs, close relatives of LPs, almost always solved through their dual — because the dual depends on the data only through inner products, which is exactly what makes the kernel trick possible. There, duality is not a convenience; it is the mechanism.

The Unsolved Core

For a theory this settled, an uncomfortable gap remains at the centre. Nobody knows whether linear programming admits a strongly polynomial algorithm — one whose count of arithmetic operations depends only on m and n , not on the bit-length of the inputs. It is problem nine on Steve Smale's list of mathematical problems for the twenty-first century, still open, with progress confined to LPs of special sparse structure [14]. Every five-minute market clearing in North America runs on algorithms whose worst case is either exponential or a function of how many digits the inputs happen to have.

The deeper strangeness is how much of this works by being invisible. Kantorovich built his multipliers to plan a command economy, and they unsettled Soviet orthodoxy because they were, mathematically, prices — computed rather than decreed. Run the same mathematics through a market instead of a planning bureau and it does not merely permit prices, it produces them: nobody at an ISO decides what a megawatt-hour in West Texas is worth at three on a windy Tuesday afternoon. It falls out of a constraint that binds. Duality says scarcity and value are the same mathematical object seen from two directions, which is a satisfying claim until you notice that it also means the price is only as meaningful as the constraint set someone wrote down. Change the model — add the integer commitment decisions back, mis-specify a line rating, leave out a physical limit the network actually has — and the number changes with it, still arriving with all the authority of something that nobody chose.

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